

Laws of Exponents — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

$$2^3 \times 2^5 = ?$$

- (a) 2^8 (b) 2^{15} (c) 4^8 (d) 2^2

Q2 • MCQ**[1]**

$$(5^2)^3 = ?$$

- (a) 5^5 (b) 5^6 (c) 5^8 (d) 5^9

Q3 • FILL IN THE BLANK**[1]**

$$7^9 \div 7^4 = 7 \text{ raised to the power:}$$

Q4 • MCQ**[1]**

$$4^3 \times 5^3 = ?$$

- (a) 20^3 (b) 20^6 (c) 20^9 (d) 9^3

Q5 • TRUE / FALSE**[1]**

$$(a^m)^n = a^{m \times n} \text{ for any base } a.$$

True False

Q6 · ASSERTION-REASON**[1]****Assertion (A):** $(5^2)^3 = 5^6$.**Reason (R):** According to the laws of exponents, $(a^m)^n = a^{m \times n}$.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)**Q7****[2]**Simplify and write in exponential form: (i) $3^4 \times 3^2$ (ii) $6^7 \div 6^3$.

Q8**[2]**Write 8^6 as a power of a power in two different ways.

Q9**[2]**Find the values of $(2^3)^2$ and $2^3 \times 2^2$, and state which is larger.

Long answer (3 marks each)**Q10****[3]**Simplify, writing each answer in exponential form: (i) $2^5 \times 2^3 \div 2^4$ (ii) $(3^2)^4 \div 3^5$.

Q11

[3]

Simplify and express as a single power: (i) $p^4 \times p^6$ (ii) $(x^5 \times x^3) \div x^2$.

Q12

[3]

A square has side 2^3 cm and a cube has edge 2^3 cm. Write the square's area and the cube's volume as powers of 2, and give their values.

Total: 21 marks · Check your work against the [answer key](#).